

Week 15 – Skill

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GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int pipefd[2];
    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }
    pid_t p1 = fork();
    if (p1 == 0) {
        int fd = open("result.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
        if (fd < 0) {
            perror("open");
            exit(1);
        }
        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[0]);
        close(pipefd[1]);
        execlp("ls", "ls", NULL);
        perror("ls");
        exit(1);
    }
    pid_t p2 = fork();
    if (p2 == 0) {
        int fd = open("result.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
        if (fd < 0) {
            perror("open");
            exit(1);
        }
        dup2(pipefd[0], STDIN_FILENO);
        dup2(fd, STDOUT_FILENO);
        close(fd);
        close(pipefd[0]);
```

```

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if (p1 == 0) {
int fd = open("result.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
if (fd < 0) {
perror("open");
exit(1);
}
dup2(pipefd[1], STDOUT_FILENO);
close(pipefd[0]);
close(pipefd[1]);
execlp("ls", "ls", NULL);
perror("ls");
exit(1);
}
pid_t p2 = fork();
if (p2 == 0) {
int fd = open("result.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
if (fd < 0) {
perror("open");
exit(1);
}
dup2(pipefd[0], STDIN_FILENO);
dup2(fd, STDOUT_FILENO);
close(fd);
close(pipefd[0]);
close(pipefd[1]);
execlp("wc", "wc", "-l", NULL);
perror("wc");
exit(1);
}
close(pipefd[0]);
close(pipefd[1]);
waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);
printf("Result stored in result.txt\n");
return 0;
}

```

```

samanth@ubuntu:~$ nano skill-15.c
samanth@ubuntu:~$ gcc skill-15.c -o skill-15
samanth@ubuntu:~$ ./skill-15
Result stored in result.txt

```